

TARUN CHOWDARY YEGI

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PROFESSIONAL SUMMARY

Computer Science undergraduate with a strong interest in applied Machine Learning, scalable system design, and security-focused applications. Experienced in developing data-driven models and full-stack systems with an emphasis on robustness, reliability, and interpretability. Comfortable working across the complete development lifecycle, from data analysis and model evaluation to backend architecture and deployment. Passionate about building intelligent systems that perform consistently under real-world constraints and evolving data conditions.

EDUCATION

Indian Institute of Engineering Science and Technology (IIST), Shibpur

Anticipated May 2028

B.Tech in Computer Science and Technology

CGPA: 8.4

Narayana Junior College

2022 – 2024

Intermediate (MPC)

96.3%

PROJECTS

Phishing URL Detection System (Random Forest-Based)

Python, Scikit-learn, SHAP, XGBoost, Pandas, Numpy ([GitHub](#))

- Developed a high-performance phishing URL detection system achieving 99%+ in-distribution accuracy across UCI and PhiUSIIL public datasets.
- Performed adversarial robustness testing including random noise perturbation (10%) and targeted structural mimic attacks, evaluating performance degradation quantitatively.
- Conducted cross-dataset evaluation (train on UCI ↔ test on PhiUSIIL) to analyze generalization under distribution shift.
- Integrated SHAP for explainable AI and feature attribution; identified SSLfinal_State, URL_of_Anchor, web_traffic, and having_Sub_Domain as dominant indicators.
- Research paper “Robustness and Explainability Analysis of ML Models for Phishing URL Detection” currently under review.

CampusWhisper – College Based Real-Time Web Chat Application

MongoDB, Express.js, React.js, Node.js (MERN Stack), WebSockets ([GitHub](#)) ([Live Site](#))

- Built a scalable real-time web chat platform using the MERN stack, enabling seamless college-specific and cross-college communication.
- Implemented an event-driven WebSocket architecture to support high-concurrency real-time messaging with low latency and efficient handling of simultaneous user sessions.
- Engineered an automatic message self-destruction mechanism (5-minute lifecycle), enforcing zero long-term data retention to enhance privacy.
- Deployed the application in a cloud environment and configured production-ready environment variables and session controls to ensure stable live web traffic handling.

EXPERIENCE

Phishing URL Detection – Robustness and Explainability Analysis

2026

Independent Research Project

Manuscript Under Review

- Conducted a comprehensive robustness analysis of phishing URL detection models by evaluating performance degradation under controlled adversarial perturbations, including stochastic noise injection and targeted structural feature manipulation.
- Designed and implemented a cross-dataset generalization framework using UCI and PhiUSIIL benchmark datasets to quantify distribution shift impact on model stability and transferability.
- Performed comparative evaluation of linear and ensemble classifiers (Logistic Regression, Random Forest, XGBoost) to analyze bias–variance trade-offs and non-linear decision boundary effectiveness in security-critical classification tasks.
- Applied SHAP-based explainability techniques to perform global feature attribution analysis, systematically identifying structural and domain-level indicators contributing to phishing detection outcomes.
- Investigated the relationship between feature redundancy, separability, and over-optimistic in-distribution performance, demonstrating the necessity of robustness-aware evaluation protocols for real-world deployment of ML-based security systems.

SKILLS

- Expertise:** Machine Learning, Exploratory Data Analysis (EDA), Adversarial Robustness Testing, Web Development, Backend & Database Management, Data Structures
- Languages:** Python, C, C++, HTML, CSS, JavaScript, SQL
- Frameworks & Libraries:** Scikit-learn, XGBoost, SHAP, Pandas, NumPy, React.js, Express.js, Node.js, MongoDB, MySQL, WebSockets, Firebase
- Tools:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Android Studio

CERTIFICATIONS & ACHIEVEMENTS

- Winter Consulting 2025 – Consulting & Business Strategy Program (IIT Guwahati). Completed 4-week intensive consulting program; Placed in Top 25 percentile. ([Certificate Link](#))
- Secured 98.9 Percentile in JEE Mains 2024 (Top 1.3% among almost 1.3 million candidates)